FINA 3103 – Homework 1, Part B

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Inputs

Elaine is Yining Wang and Chen is Ziyou Chen. The certainty equivalents (CEs) from Part A are used exactly as submitted.

Table 1: Certainty equivalents from Part A

Bet	Outcomes	Erik	Elaine	Chen
X	50%: +30%, 50%: −10%	10%	4%	7%
Y	80%: +15%, 20%: −25%	12%	3%	5%
Z	20%: +39%, 80%: −1%	5%	3.5%	2%

Table 2: Capital market

Asset	Expected return	Standard deviation
Stock fund S	10%	18%
Bond fund B	6%	10%
Risk-free	3%	0%
Correlation ρ_{SB}	0.25	
Borrowing rate r_{borrow} (part c)	6%	

(a) Risk aversion from the certainty equivalents

For a bet with outcomes r_i and probabilities p_i:

$$E[r] = \sum_i p_i r_i, \qquad \sigma^2 = \sum_i p_i \big(r_i - E[r]\big)^2, \qquad \text{skew} = \frac{\sum_i p_i \big(r_i - E[r]\big)^3}{\sigma^3}.$$

Under mean-variance utility the CE equals the utility of the bet:

$$CE = E[r] - \tfrac{1}{2} A \sigma^2 \quad \Longrightarrow \quad A = \frac{2\big(E[r] - CE\big)}{\sigma^2}.$$

Table 3: Moments of the bets

Bet	$E[r]$	σ	σ^2	Skewness
X	10%	20%	0.0400	0.00
Y	7%	16%	0.0256	−1.50
Z	7%	16%	0.0256	+1.50

Table 4: Implied risk aversion A

Member	Bet X	Bet Y	Bet Z
Erik	0.000	−3.906	1.563
Elaine	3.000	3.125	2.734
Chen	1.500	1.563	3.906

What should hold for Bets Y and Z. Y and Z have the same expected return and the same variance (Table 3). A mean-variance investor cares only about these two moments, so each member should state the *same* CE for Y and Z, and the implied A should be the same for every bet.

Does it hold for our group? No. Every member gives different CEs for Y and Z (Table 1), so the implied A differs across bets (Table 4).

- **Erik** is risk-neutral on X, risk-loving on Y (his CE is above the expected return, so $A < 0$) and risk-averse on Z.
- **Elaine** has a fairly stable A across the three bets, so mean-variance utility is a reasonable approximation for her.
- **Chen**'s A for Z is more than twice his A for Y.

Which assumption this contradicts. Y and Z differ only in their skewness (Table 3): Y has a small chance of a large loss (negative skew), Z a small chance of a large gain (positive skew). Since our CEs differ between them, we care about more than the mean and the variance. This contradicts the assumption that returns are (approximately) normally distributed, or equivalently that investors care only about the first two moments. The variation of A across bets also contradicts a single, constant risk-aversion coefficient per investor. Elaine values the positively skewed Z more (the usual preference for skewness), while Erik and Chen prefer the negatively skewed Y.

(b) Optimal risky portfolio and complete portfolios

The risky portfolio P^*. Every investor chooses the risky portfolio with the steepest capital allocation line (CAL), i.e. the highest Sharpe ratio:

$$\max_{w_S} S_P = \frac{E[r_P] - r_f}{\sigma_P}, \qquad E[r_P] = w_S E[r_S] + w_B E[r_B], \qquad \sigma_P^2 = w_S^2\sigma_S^2 + w_B^2\sigma_B^2 + 2w_S w_B \sigma_{SB}.$$

The first-order condition gives $\mathbf{w}^* \propto \Sigma^{-1}(\boldsymbol{\mu} - r_f\mathbf{1})$. With excess returns $R_S = E[r_S] - r_f$, $R_B = E[r_B] - r_f$ and $\sigma_{SB} = \rho_{SB}\sigma_S\sigma_B$, this becomes for two assets

$$w_S^* = \frac{R_S\,\sigma_B^2 - R_B\,\sigma_{SB}}{R_S\,\sigma_B^2 + R_B\,\sigma_S^2 - (R_S + R_B)\,\sigma_{SB}}, \qquad w_B^* = 1 - w_S^*.$$

Table 5: Inputs to the tangency formula

R_S	R_B	σ_S^2	σ_B^2	σ_{SB}	Numerator	Denominator
0.07	0.03	0.0324	0.0100	0.0045	0.000565	0.001222

Table 6: Optimal risky portfolio P^*

w_S^*	w_B^*	w_S^*/w_B^*	$E[r_{P^*}]$	σ_{P^*}	Sharpe ratio
0.4624	0.5376	0.8600	7.85%	10.98%	0.4417

Capital allocation. The complete portfolio C puts a fraction y in P^* and $1-y$ in the risk-free asset:

$$E[r_C] = r_f + y\big(E[r_{P^*}] - r_f\big), \qquad \sigma_C = y\,\sigma_{P^*}.$$

Maximising $U(y) = E[r_C] - \frac{1}{2}A\sigma_C^2$:

$$U(y) = r_f + y\big(E[r_{P^*}] - r_f\big) - \tfrac{1}{2}Ay^2\sigma_{P^*}^2,$$
$$\frac{dU}{dy} = \big(E[r_{P^*}] - r_f\big) - Ay\,\sigma_{P^*}^2 = 0 \quad\Longrightarrow\quad y^* = \frac{E[r_{P^*}] - r_f}{A\,\sigma_{P^*}^2}.$$

The complete-portfolio weights are $\big(y^*w_S^*,\ y^*w_B^*,\ 1-y^*\big)$. We use each member's A from Bet X.

Table 7: Complete portfolios with one rate for borrowing and lending

Member	A_X	y^*	w_S	w_B	w_{r_f}	w_S/w_B	$E[r_C]$	σ_C	U
Erik	0.00	∞	∞	∞	$-\infty$	0.8600	–	–	–
Elaine	3.00	1.3410	0.6200	0.7210	−0.3410	0.8600	9.50%	14.72%	0.0625
Chen	1.50	2.6820	1.2401	1.4420	−1.6820	0.8600	16.01%	29.45%	0.0950

Erik's CE for Bet X equals its expected return, so $A_X = 0$: he is risk-neutral, y^* is unbounded and he would borrow without limit at r_f to buy P^*. Elaine and Chen both have $y^* > 1$ and borrow at the risk-free rate.

Same ratio of S to B. The column w_S/w_B in Table 7 is identical for every member and equals w_S^*/w_B^* in Table 6. Risk aversion only changes how much is invested in P^*, never the mix of S and B inside it. This is the *separation property* (two-fund separation): the optimal risky portfolio is independent of preferences, and A only determines the allocation between P^* and the risk-free asset.

(c) Borrowing at 6%, lending at 3%

Lending versus borrowing. The sign of the risk-free weight $1-y$ determines the side:

- **Lending** ($y \le 1$, $w_{r_f} \ge 0$): part of wealth is invested in the risk-free asset and earns r_f.
- **Borrowing** ($y > 1$, $w_{r_f} < 0$): the investor holds more than 100% in P^* and finances the excess $y-1$ with a loan at r_{borrow}.

The higher rate applies only to the borrowed amount, so lenders are unaffected. The CAL gets a kink at P^*:

$$E[r_C] = \begin{cases} r_f + y\big(E[r_{P^*}] - r_f\big), & y \le 1, \\ r_{\text{borrow}} + y\big(E[r_{P^*}] - r_{\text{borrow}}\big), & y > 1, \end{cases} \qquad \sigma_C = y\,\sigma_{P^*},$$

$$\text{slope of the CAL} = \begin{cases} \dfrac{E[r_{P^*}] - r_f}{\sigma_{P^*}}, & y \le 1, \\ \dfrac{E[r_{P^*}] - r_{\text{borrow}}}{\sigma_{P^*}}, & y > 1. \end{cases}$$

The first-order condition on each segment gives

$$y_L = \frac{E[r_{P^*}] - r_f}{A\sigma_{P^*}^2}, \qquad y_B = \frac{E[r_{P^*}] - r_{\text{borrow}}}{A\sigma_{P^*}^2}, \qquad y^* = \begin{cases} y_L & \text{if } y_L \le 1 \text{ (lends, unaffected)}, \\ 1 & \text{if } y_B < 1 < y_L \text{ (at the kink)}, \\ y_B & \text{if } y_B \ge 1 \text{ (borrows at } r_{\text{borrow}}). \end{cases}$$

Utility is evaluated on the segment the investor ends up on: $U = E[r_C] - \frac{1}{2}Ay^{*2}\sigma_{P^*}^2$.

Range of A for holding exactly 100% in P^*. The investor neither borrows nor lends when $y_B \le 1 \le y_L$:

$$\frac{E[r_{P^*}] - r_{\text{borrow}}}{\sigma_{P^*}^2} \;\le\; A \;\le\; \frac{E[r_{P^*}] - r_f}{\sigma_{P^*}^2}.$$

Table 8: The kinked CAL

Slope, lending	Slope, borrowing	Lower bound on A	Upper bound on A
0.4417	0.1684	1.5343	4.0230

Table 9: Complete portfolios with $r_{\text{borrow}} = 6\%$

Member	A_X	y_L	y_B	y^*	w_S	w_B	w_{r_f}	$E[r_C]$	σ_C	U
Erik	0.00	∞	∞	∞	∞	∞	$-\infty$	–	–	–
Elaine	3.00	1.3410	0.5114	1.0000	0.4624	0.5376	0.0000	7.85%	10.98%	0.0604
Chen	1.50	2.6820	1.0228	1.0228	0.4729	0.5499	−0.0228	7.89%	11.23%	0.0695

Table 10: Effect of the borrowing rate

Member	y^* before	y^* after	w_{r_f} before	w_{r_f} after	U before	U after
Erik	∞	∞	$-\infty$	$-\infty$	–	–
Elaine	1.3410	1.0000	−0.3410	0.0000	0.0625	0.0604
Chen	2.6820	1.0228	−1.6820	−0.0228	0.0950	0.0695

Who is affected. All three members borrowed in (b), so all three are affected.

- **Elaine** would borrow at r_f ($y_L > 1$) but not at r_{borrow} ($y_B < 1$). Her A lies inside the range in Table 8, so she ends up exactly at the kink with 100% in P^*, and her utility falls.

- **Chen**'s A lies just below the range in Table 8. He still borrows, but only a small fraction of his wealth instead of more than his whole wealth, and his utility falls.
- **Erik** is risk-neutral. Since $E[r_{P^*}] > r_{\text{borrow}}$, borrowing is still profitable in expectation and his optimal leverage remains unbounded.

(d) Should more risk-averse investors hold more bonds?

Consistency with (b). No. In (b) every member holds S and B in the same ratio (Table 7), and risk aversion only changes y^*. Under mean-variance utility with one rate for borrowing and lending, a more risk-averse investor should hold more *cash*, not more bonds. The advisers' rule contradicts the separation property.

Reason 1: Borrowing is more expensive than lending (our numbers from (c)). Separation requires that investors can borrow and lend at the same rate. With $r_{\text{borrow}} > r_f$ the CAL beyond P^* is much flatter (Table 8). An investor who wants more risk than P^* can instead get it without borrowing, by tilting the risky portfolio towards stocks. Maximising $E[r_P] - \frac{1}{2}A\sigma_P^2$ over the stock-bond mix with $y = 1$ gives

$$w_S = \frac{\big(E[r_S] - E[r_B]\big)/A + \sigma_B^2 - \sigma_{SB}}{\sigma_S^2 + \sigma_B^2 - 2\sigma_{SB}}, \qquad w_B = 1 - w_S,$$

which is optimal for $0.7243 \le A \le 4.0230$, i.e. between the tangency portfolio relative to r_{borrow} and P^*.

Table 11: Holding P^* (from (c)) versus choosing the stock-bond mix freely

	P^* fixed, part (c)				Mix chosen freely, $y = 1$			
Member	w_S	w_B	w_{r_f}	U	w_S	w_B	w_{r_f}	U
Elaine ($A = 3$)	0.4624	0.5376	0.0000	0.0604	0.5639	0.4361	0.0000	0.0609
Chen ($A = 1.5$)	0.4729	0.5499	−0.0228	0.0695	0.9631	0.0369	0.0000	0.0757

Both are better off leaving P^*, and the optimal mix now depends on A: Chen, the less risk-averse, holds almost only stocks, while Elaine holds a larger share of bonds. Investors with A above the upper bound in Table 8 lend and keep P^*. With a borrowing spread the advisers' rule is therefore rational.

Reason 2: Preferences depend on more than mean and variance (our numbers from (a)). Bets Y and Z have identical mean and variance, yet every member gives them different CEs:

Table 12: CEs and implied A for the two bets with equal mean and variance

Member	CE(Y)	CE(Z)	A_Y	A_Z
Erik	12%	5%	−3.906	1.563
Elaine	3%	3.5%	3.125	2.734
Chen	5%	2%	1.563	3.906

We care about skewness and tail risk, and P^* is optimal only for an investor who looks at the first two moments. An investor who dislikes left-tail risk, as Elaine does by valuing Z above Y, would rationally hold less of the asset with larger crash risk (typically stocks) and more bonds than in P^*.

Other reasons. An investor with future liabilities or a long horizon may hold long-term bonds as a hedge, since they are close to risk-free for that horizon; labour income that is correlated with the stock market makes stocks riskier for that investor; and estimation error in expected returns makes the mean-variance weights unreliable in practice.